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Next item →

1. What was the critical assumption that we made when finding an equation to express $\mathbb{E}[\tilde{x}_k^- \mid z_k]$?

1 / 1 point

- ☐ That $\tilde{x}_k^-$ and z_k are both scalar random variables.
- ☐ That the state equation of the model, which updates x_k every time step, is a linear equation.
- ☐ That the output equation of the model, which computes z_k every time step, is a linear equation.
- ☒ That $\tilde{x}_k^-$ and z_k have a joint Gaussian distribution.
- ☐ That $\tilde{x}_k^-$ and z_k are uncorrelated.

✔ Correct
Yes. That allows us to compute the conditional expectation.

2. If X and Z are jointly Gaussian random vectors and z is a constant known vector, what is $\mathbb{E}[X \mid Z = z]$?

1 / 1 point

- ☒ $\mathbb{E}[X \mid Z = z] = \mathbb{E}[X] + \Sigma_{\tilde{X}\tilde{Z}}\Sigma_{\tilde{Z}}^{-1}(z - \mathbb{E}[Z]).$
- ☐ $\mathbb{E}[X \mid Z = z] = \mathbb{E}[Z].$
- ☐ $\mathbb{E}[X \mid Z = z] = \Sigma_{\tilde{X}} - \Sigma_{\tilde{X}\tilde{Z}}\Sigma_{\tilde{Z}}^{-1}\Sigma_{\tilde{Z}\tilde{X}}^T.$
- ☐ $\mathbb{E}[X \mid Z = z] = \mathbb{E}[X].$
- ☐ $\mathbb{E}[X \mid Z = z] = \frac{\mathbb{E}[X] + \mathbb{E}[Z]}{2}.$

✔ Correct
Yes. This is one of the main results of this lesson.

3. What are the uses of the Kalman-filter gain factor L_k ?

1 / 1 point

- ☐ It is used in a Kalman-filtering step to eliminate noise from the measured output when updating the state estimate.
- ☐ It is a gain factor used to maximize the covariance $\Sigma_{\hat{x},k}^+.$
- ☒ It is a scale factor used in computing the uncertainty of the state estimate from the uncertainty of the prediction.

✔ Correct
Yes. This is one use of the gain factor.

- ☒ It is the optimal blend factor between new information in $\tilde{z}_k$ and the prediction $\hat{x}_k^-$ when computing the state estimate $\hat{x}_k^+.$

✔ Correct
Yes. This is its main purpose.

- ☐ It is the reciprocal of the Schur complement of a block covariance matrix used in the derivation.